Task 02: Infrastructure & Technology Profiling

I performed technology and infrastructure profiling on the target website using WhatWeb to identify technologies such as JavaScript libraries, HTTP headers, cookies, and server configurations. I also used WHOIS and nslookup to gather domain information and analyze DNS records including A, MX, NS, and TXT. Additionally, advanced Google dorking on Google Search was used to discover publicly exposed files like .env, backup archives, logs, database files, and robots.txt, helping identify potential security misconfigurations.

1. WHOIS

Purpose: WHOIS is used to find information about a domain such as:

- Domain owner
- Registrar
- IP ranges
- Name servers
- Registration dates

Command:

```
whois hackthissite.org
```

What it tells you:

- Organization owning the domain
- Name servers (DNS servers)
- IP blocks sometimes belonging to the company
- Contact information

Why pentesters use it:

Helps identify infrastructure owned by the organization and sometimes reveals additional domains.

POC:

```
┌──(hyena㉿hyena)-[~/Downloads]
└─$ whatweb -v -a 3 https://hackthissite.org

WhatWeb report for https://hackthissite.org
Status    : 200 OK
Title     : Hack This Site
IP        : 137.74.187.104
Country   : CANADA, CA

Summary   : Content-Language[en], Cookies[HackThisSite], HTTPServer[HackThisSite], JQuery[1.8.1], Meta-Author[HackThisSite.org Staff],  Open-Graph-Protocol[we
bsite], PasswordField[password], Script[text/javascript], Strict-Transport-Security[max-age=31536000; includeSubDomains; preload], UncommonHeaders[upgrade,on
ion-location,access-control-allow-origin,content-security-policy,referrer-policy,feature-policy,public-key-pins-report-only,report-to,nel], X-XSS-Protection[
0]

Detected Plugins:
[ Content-Language ]
        Detect the content-language setting from the HTTP header.

        String       : en

[ Cookies ]
        Display the names of cookies in the HTTP headers. The
        values are not returned to save on space.

        String       : HackThisSite

[ HTTPServer ]
        HTTP server header string. This plugin also attempts to
        identify the operating system from the server header.

        String       : HackThisSite (from server string)

[ JQuery ]
        A fast, concise, JavaScript that simplifies how to traverse
        HTML documents, handle events, perform animations, and add
        AJAX.

        Version      : 1.8.1
        Website     : http://jquery.com/

[ Meta-Author ]
        This plugin retrieves the author name from the meta name
        tag - info:
        http://www.webmarketingnow.com/tips/meta-tags-uncovered.html
        #author

        String       : HackThisSite.org Staff

[ Open-Graph-Protocol ]
        The Open Graph protocol enables you to integrate your Web
```

```
        pages into the social graph. It is currently designed for
        Web pages representing profiles of real-world things .
        things like movies, sports teams, celebrities, and
        restaurants. Including Open Graph tags on your Web page,
        makes your page equivalent to a Facebook Page.

        Version      : website

[ PasswordField ]
        find password fields

        String       : password (from field name)

[ Script ]
        This plugin detects instances of script HTML elements and
        returns the script language/type.

        String       : text/javascript

[ Strict-Transport-Security ]
        Strict-Transport-Security is an HTTP header that restricts
        a web browser from accessing a website without the security
        of the HTTPS protocol.

        String       : max-age=31536000; includeSubDomains; preload

[ UncommonHeaders ]
        Uncommon HTTP server headers. The blacklist includes all
        the standard headers and many non standard but common ones.
        Interesting but fairly common headers should have their own
        plugins, eg. x-powered-by, server and x-aspnet-version.
        Info about headers can be found at www.http-stats.com

        String       : upgrade,onion-location,access-control-allow-origin,content-security-policy,referrer-policy,feature-policy,public-key-pins-report-only,
report-to,nel (from headers)

[ X-XSS-Protection ]
        This plugin retrieves the X-XSS-Protection value from the
        HTTP header. - More Info:
        http://msdn.microsoft.com/en-us/library/cc288472%28VS.85%29.
        aspx

        String       : 0

HTTP Headers:
        HTTP/1.1 200 OK
        Date: Thu, 05 Mar 2026 17:08:09 GMT
        Upgrade: h2,h2c
        Connection: Upgrade, close
        Set-Cookie: HackThisSite=2hnrnfq5greo0971n0i51u73g1; expires=Fri, 06-Mar-2026 17:08:09 GMT; path=/
        Expires: Thu, 19 Nov 1981 08:52:00 GMT
```

```
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        the standard headers and many non standard but common ones.
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HTTP Headers:
        HTTP/1.1 200 OK
        Date: Thu, 05 Mar 2026 17:08:09 GMT
        Upgrade: h2,h2c
        Connection: Upgrade, close
        Set-Cookie: HackThisSite=2hnrnfq5greo097ln0i5lu73gl; expires=Fri, 06-Mar-2026 17:08:09 GMT; path=/
        Expires: Thu, 19 Nov 1981 08:52:00 GMT
        Cache-Control: no-store, no-cache, must-revalidate, post-check=0, pre-check=0
        Pragma: no-cache
        Onion-Location: http://hackthisjogneh42n5o7gbzrewxee3vyu6ex37ukyvdw6jm66npakiyd.onion/
        Vary: Accept-Encoding
        Content-Encoding: gzip
        Content-Length: 6502
        Content-Type: text/html
        Content-Language: en
        Server: HackThisSite
        Access-Control-Allow-Origin: *
        Content-Security-Policy: child-src 'self' hackthissite.org *.hackthissite.org htscdn.org *.htscdn.org discord.com; form-action 'self' hackthissite.or
g *.hackthissite.org htscdn.org *.htscdn.org; upgrade-insecure-requests; report-uri https://hackthissite.report-uri.com/r/d/csp/enforce
        Referrer-Policy: origin-when-cross-origin
        X-XSS-Protection: 0
        Feature-Policy: fullscreen *
        Public-Key-Pins-Report-Only: pin-sha256="YLh1dUR9y6Kja30RrAn7JKnbQG/uEtLMkBgFF2Fuihg="; pin-sha256="Vjs8r4z+80wjNcr1YKepWQboSIRi63WsWXhIMN+eWys="; ma
x-age=2592000; includeSubDomains; report-uri="https://hackthissite.report-uri.com/r/d/hpkp/reportOnly"
        Strict-Transport-Security: max-age=31536000; includeSubDomains; preload
        Report-To: {"group":"default","max_age":31536000,"endpoints":[{"url":"https://hackthissite.report-uri.com/a/d/g"}],"include_subdomains":true}
        NEL: {"report_to":"default","max_age":31536000,"include_subdomains":true,"success_fraction":0.0,"failure_fraction":0.1}
```

2. NSLOOKUP

Purpose:NSLookup is used to query DNS records of a domain.

Basic Command:

nslookup hackthissite.org

Output shows:

- Domain IP address
- DNS server used to resolve the query

Common DNS Records to Query

1. Check A Record (IP address)

   nslookup hackthissite.org

2. Check Name Servers

   nslookup -type=ns hackthissite.org

3. Check Mail Servers

   nslookup -type=mx hackthissite.org

4. Check TXT Records (sometimes contain secrets)

nslookup -type=txt hackthissite.org

Example Workflow in Recon

1. Find domain info
whois hackthissite.org
2. Find domain IP
nslookup hackthissite.org
3. Find name servers
nslookup -type=ns hackthissite.org
4. Find mail servers
nslookup -type=mx hackthissite.org

Why this is important

- Reveals infrastructure
- Shows DNS configuration
- Helps find additional attack surface
- Sometimes exposes internal services

DNS reconnaissance is important because it reveals the target's infrastructure, servers, email systems, and possible weak points before scanning or attacking.

```
┌──(hyena㉿hyena)-[~]
└─$ nslookup hackthissite.org
erver:         192.168.1.1
ddress:        192.168.1.1#53

on-authoritative answer:
ame:   hackthissite.org
ddress: 137.74.187.101
ame:   hackthissite.org
ddress: 137.74.187.100
ame:   hackthissite.org
ddress: 137.74.187.102
ame:   hackthissite.org
ddress: 137.74.187.104
ame:   hackthissite.org
ddress: 137.74.187.103


┌──(hyena㉿hyena)-[~]
```

```
┌──(hyena㉿hyena)-[~]
└─$ nslookup -type=ns hackthissite.org
Server:         192.168.1.1
Address:        192.168.1.1#53

Non-authoritative answer:
hackthissite.org        nameserver = h.ns.buddyns.com.
hackthissite.org        nameserver = g.ns.buddyns.com.
hackthissite.org        nameserver = j.ns.buddyns.com.
hackthissite.org        nameserver = f.ns.buddyns.com.
hackthissite.org        nameserver = c.ns.buddyns.com.

Authoritative answers can be found from:
f.ns.buddyns.com        internet address = 5.223.55.119
f.ns.buddyns.com        has AAAA address 2a01:4ff:2f0:3661::3

┌──(hyena㉿hyena)-[~]
└─$ nslookup -type=mx hackthissite.org
Server:         192.168.1.1
Address:        192.168.1.1#53

Non-authoritative answer:
hackthissite.org        mail exchanger = 30 aspmx5.googlemail.com.
hackthissite.org        mail exchanger = 30 aspmx2.googlemail.com.
hackthissite.org        mail exchanger = 20 alt1.aspmx.l.google.com.
hackthissite.org        mail exchanger = 10 aspmx.l.google.com.
hackthissite.org        mail exchanger = 30 aspmx3.googlemail.com.
hackthissite.org        mail exchanger = 20 alt2.aspmx.l.google.com.
hackthissite.org        mail exchanger = 30 aspmx4.googlemail.com.

Authoritative answers can be found from:
```

```
┌──(hyena㉿hyena)-[~]
└─$ whois hackthissite.org
Domain Name: hackthissite.org
Registry Domain ID: REDACTED
Registrar WHOIS Server: http://whois.porkbun.com
Registrar URL: https://porkbun.com
Updated Date: 2025-07-15T23:08:30Z
Creation Date: 2003-08-10T15:01:25Z
Registry Expiry Date: 2026-08-10T15:01:25Z
Registrar: Porkbun LLC
Registrar IANA ID: 1861
Registrar Abuse Contact Email: abuse@porkbun.com
Registrar Abuse Contact Phone: +1.8557675286
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: c.ns.buddyns.com
Name Server: f.ns.buddyns.com
Name Server: g.ns.buddyns.com
Name Server: h.ns.buddyns.com
Name Server: j.ns.buddyns.com
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://icann.org/wicf/
>>> Last update of WHOIS database: 2026-03-05T17:33:34Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

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cord in the Public Interest Registry registry database. The data in this record is provided by Public Interest Registry for informational purposes only, and
Public Interest Registry does not guarantee its accuracy. This service is intended only for query-based access. You agree that you will use this data only fo
r lawful purposes and that, under no circumstances will you use this data to (a) allow, enable, or otherwise support the transmission by e-mail, telephone, o
r facsimile of mass unsolicited, commercial advertising or solicitations to entities other than the data recipient's own existing customers; or (b) enable hi
gh volume, automated, electronic processes that send queries or data to the systems of Registry Operator, a Registrar, or Identity Digital except as reasonab
ly necessary to register domain names or modify existing registrations. All rights reserved. Public Interest Registry reserves the right to modify these term
s at any time. By submitting this query, you agree to abide by this policy.  The Registrar of Record identified in this output may have an RDDS service that
can be queried for additional information on how to contact the Registrant, Admin, or Tech contact of the queried domain name.
```

4.WhatWeb

Purpose: Website fingerprinting tool used to detect technologies like CMS, frameworks, server type, and libraries used by a website.

Command:

whatweb -v -a 3 https://hackthissite.org

Explanation:

what web

Website fingerprinting tool used to detect technologies like CMS, frameworks, server type, and libraries used by a website.

-v

Verbose mode.

Shows detailed output like detected plugins, headers, and technology information.

-a 3

Aggression level.

Controls how deeply WhatWeb scans the target.

Levels:

-a 1 → Passive scan

-a 2 → Moderate scan

-a 3 → Aggressive scan (commonly used)

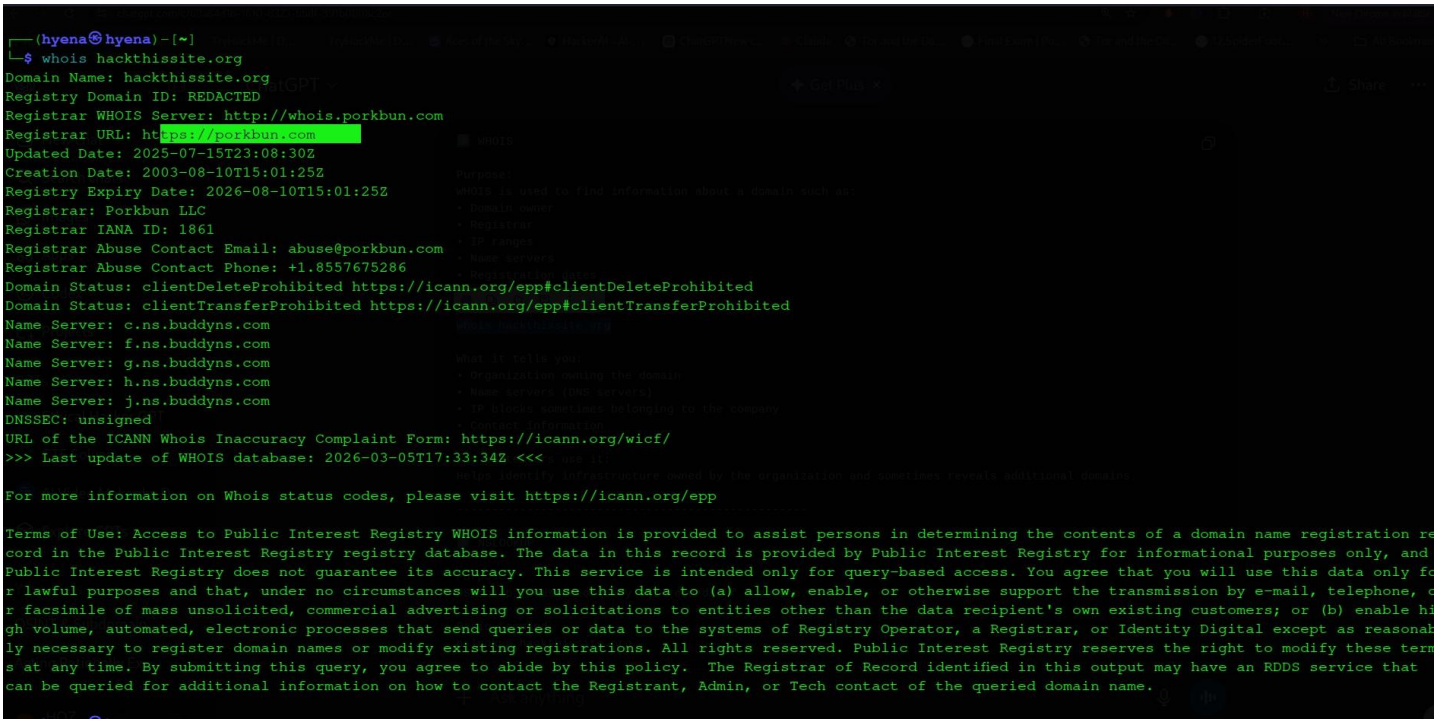
-a 4 → Very aggressive scan

https://hackthissite.org
Target website being scanned.

Most Useful Commands:

1. Scan a single website
   whatweb https://example.com
2. Scan with aggressive detection
   whatweb -a 3 https://example.com
3. Scan multiple targets from file
   whatweb -i targets.txt
4. Save results to file
   whatweb -i targets.txt --log-brief results.txt
5. Detailed output report
   whatweb -i targets.txt --log-verbose detailed.txt

POC:



```
┌──(hyena㉿hyena)-[~]
└─$ whois hackthissite.org
Domain Name: hackthissite.org
Registry Domain ID: REDACTED
Registrar WHOIS Server: http://whois.porkbun.com
Registrar URL: https://porkbun.com
Updated Date: 2025-07-15T23:08:30Z
Creation Date: 2003-08-10T15:01:25Z
Registry Expiry Date: 2026-08-10T15:01:25Z
Registrar: Porkbun LLC
Registrar IANA ID: 1861
Registrar Abuse Contact Email: abuse@porkbun.com
Registrar Abuse Contact Phone: +1.8557675286
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: c.ns.buddyns.com
Name Server: f.ns.buddyns.com
Name Server: g.ns.buddyns.com
Name Server: h.ns.buddyns.com
Name Server: j.ns.buddyns.com
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://icann.org/wicf/
>>> Last update of WHOIS database: 2026-03-05T17:33:34Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

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Public Interest Registry does not guarantee its accuracy. This service is intended only for query-based access. You agree that you will use this data only fo
r lawful purposes and that, under no circumstances will you use this data to (a) allow, enable, or otherwise support the transmission by e-mail, telephone, o
r facsimile of mass unsolicited, commercial advertising or solicitations to entities other than the data recipient's own existing customers; or (b) enable hi
gh volume, automated, electronic processes that send queries or data to the systems of Registry Operator, a Registrar, or Identity Digital except as reasonab
ly necessary to register domain names or modify existing registrations. All rights reserved. Public Interest Registry reserves the right to modify these term
s at any time. By submitting this query, you agree to abide by this policy.  The Registrar of Record identified in this output may have an RDDS service that
can be queried for additional information on how to contact the Registrant, Admin, or Tech contact of the queried domain name.
```

4.Google Dorking

Google Dorking is an advanced search technique that uses special operators in Google Search to

find sensitive information, hidden pages, or misconfigured files that are indexed by search engines. It helps security researchers discover exposed resources such as configuration files, backup archives, login panels, and debugging pages.

Purpose:
To identify publicly accessible files, directories, and sensitive information related to the target domain.

Common Google Dorking Commands Used:

site:hackthissite.org

site:*.hackthissite.org

site:hackthissite.org inurl:admin

site:hackthissite.org inurl:login

site:hackthissite.org inurl:dashboard

site:hackthissite.org inurl:config

site:hackthissite.org inurl:backup

site:hackthissite.org inurl:phpinfo

site:hackthissite.org filetype:env

site:hackthissite.org filetype:sql

site:hackthissite.org filetype:log

site:hackthissite.org filetype:zip

site:hackthissite.org filetype:bak

site:hackthissite.org filetype:json

site:hackthissite.org filetype:xml

site:hackthissite.org filetype:txt

site:hackthissite.org filetype:conf

site:hackthissite.org filetype:config

site:hackthissite.org intitle:"index of"

site:hackthissite.org intitle:"index of" backup

site:hackthissite.org intitle:"index of" uploads

site:hackthissite.org intitle:"index of" database

site:hackthissite.org "phpinfo()"

site:hackthissite.org "DB_PASSWORD"

site:hackthissite.org "api_key"

site:hackthissite.org "secret_key"

site:hackthissite.org "access_key"

site:hackthissite.org "aws_access_key"

site:hackthissite.org intext:"password"

site:hackthissite.org intext:"username"

site:hackthissite.org robots.txt

cache:hackthissite.org